

Robust State Estimation in Redundant Sensor Networks under Controlled Sensor Faults

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Abstract

Robust state estimation in sensor networks is often complicated by probe miscalibration, asynchronous sampling, and sensor faults. While calibration can enforce internal consistency among redundant probes, it does not necessarily guarantee robustness under structured failures. In this work, we present a controlled experimental study of estimation robustness in a synthetic multi-sensor environment with redundant probes and known latent ground-truth states.

We generate asynchronous sensor logs with unknown probe-level gain and bias, and evaluate baseline averaging and calibration-only estimators under nominal, benign fault, and adversarial probe-level fault conditions. To reveal localized failure modes that are obscured by global accuracy metrics, we introduce a localized fault penalty that compares estimation error during fault windows against fault-free periods.

Our results show that calibration substantially reduces probe disagreement under nominal conditions but can catastrophically fail under adversarial faults that violate probe consistency assumptions. To mitigate this failure, we propose a deterministic graph-based temporal smoothing extension that incorporates spatial context from neighboring sensors. We demonstrate that conservative spatial coupling significantly improves robustness, while excessive diffusion introduces bias due to asynchronous dynamics.

This study highlights the nuanced interaction between calibration, fault structure, and spatial reasoning in sensor networks, and establishes a principled baseline for future fault-aware and learning-based robustness methods.

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1 Introduction

Distributed sensor networks are widely used to monitor physical environments such as agricultural fields, industrial systems, and smart infrastructure. In these settings, sensors often employ redundant probes to improve reliability and reduce noise. However, probe miscalibration, asynchronous data arrival, and sensor faults remain fundamental challenges for robust state estimation.

Calibration is a commonly employed strategy to address probe misalignment by estimating relative gains and biases across redundant measurements. When probes observe the same latent state, calibration can enforce internal consistency and reduce measurement variance. As a result, calibration is often implicitly treated as a proxy for robustness. However, consistency among probes does not guarantee correctness of the latent state estimate, particularly in the presence of structured or adversarial sensor faults.

Most prior work evaluates estimation performance using global accuracy metrics such as root mean squared error (RMSE). While useful, these metrics can obscure localized failures when faults affect only a subset of sensors or limited time intervals. As a consequence, estimators may appear robust on average while exhibiting severe degradation under targeted faults.

In this work, we conduct a controlled experimental analysis to disentangle the effects of calibration, fault structure, and spatial context on estimation robustness. Using synthetically generated sensor logs with known ground truth, we evaluate baseline averaging and calibration-only estimators under nominal conditions, benign bias faults, and adversarial probe-level faults. To explicitly capture robustness under localized failures, we introduce a localized fault penalty that compares estimation error during fault windows to error during fault-free periods on a per-node basis.

Our experiments reveal that calibration, while effective at reducing probe disagreement under nominal conditions, can amplify structured probe-level failures and lead to catastrophic estimation error under adversarial faults. To address this limitation, we introduce a simple graph-based temporal smoothing extension that incorporates spatial context from neighboring sensors without learning or adaptation. We show that conservative spatial coupling can suppress fault amplification and restore robustness, while overly aggressive diffusion introduces bias due to asynchronous sensor dynamics.

By isolating estimator behavior in a controlled setting, this work provides clear empirical insights into when calibration improves robustness and when it fails. The results establish a principled foundation for future extensions involving adaptive graph-based models and explicit fault detection mechanisms.

2 Synthetic Data Generation

To enable controlled and reproducible analysis of estimator robustness, we construct a synthetic sensor network that captures key characteristics of real-world environmental monitoring systems, including redundant probes, unknown calibration errors, asynchronous sampling, and stochastic dynamics. The use of synthetic data allows full access to latent ground truth while ensuring that estimation algorithms operate solely on corrupted sensor measurements.

2.1 Sensor Network Structure

The simulated network consists of five soil sensor nodes and one pressure sensor node. Each soil node measures two physical modalities: soil moisture and temperature. For each modality, two redundant probes are deployed, resulting in four measurements per soil node at each sampling instance. The pressure node provides a single pressure measurement. This configuration reflects common redundancy patterns in deployed sensor systems, where multiple probes are used to improve reliability.

2.2 Latent Physical States

Each sensor node is associated with a latent physical state that evolves over time. Soil moisture and temperature states follow slowly varying stochastic dynamics, with moisture additionally influenced by intermittent rainfall events. Pressure evolves independently according to a noisy temporal process. These latent states represent the true environmental conditions and are never exposed to the estimation algorithms. They are retained solely for post hoc evaluation of estimator performance.

2.3 Calibration Model

To model realistic sensor imperfections, each probe is assigned an unknown affine calibration error characterized by a gain and a bias parameter. Calibration parameters are independently sampled for every probe and remain fixed throughout the experiment. The resulting measurements therefore exhibit systematic discrepancies even when observing the same latent state. Importantly, these calibration parameters are unknown to all estimators and must be inferred implicitly through probe consistency.

2.4 Measurement Process

Sensor measurements are generated asynchronously, with each node emitting packets at irregular intervals. Observations are corrupted by additive measurement noise and logged as raw text-based packets that mimic real telemetry streams. This design preserves realistic challenges such as missing observations, uneven sampling rates, and heterogeneous sensor availability across time windows.

Overall, this synthetic data generation framework provides a controlled yet realistic testbed for evaluating the robustness of state estimation methods under calibration uncertainty and sensor faults.

3 Data Parsing and Temporal Aggregation

Raw sensor logs are parsed using regular expressions and converted into a canonical long-format representation. Asynchronous measurements are aggregated into fixed-duration time windows without interpolation. Missing data is preserved to reflect realistic sensor behavior.

4 Estimators

4.1 Baseline Estimator

The baseline estimator averages redundant probes and applies exponential moving average (EMA) smoothing over time. It does not perform calibration or exploit spatial information.

4.2 Calibration-Only Estimator

The calibration-only estimator fits an affine calibration model for each probe using least-squares estimation. Calibrated measurements are inverted and averaged to obtain latent state estimates. Calibration parameters are estimated once and held fixed in all subsequent experiments.

5 Evaluation Metrics

To evaluate estimator performance under nominal and faulty conditions, we employ a set of complementary metrics that quantify both estimation accuracy and robustness. All metrics are

computed with respect to latent ground-truth states, which are used exclusively for evaluation and never exposed to the estimators.

5.1 Root Mean Squared Error

Root Mean Squared Error (RMSE) is used as the primary measure of estimation accuracy under nominal conditions. For a given modality and sensor node, RMSE is computed as

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{\theta}_t - \theta_t)^2}, \quad (1)$$

where $\hat{\theta}_t$ denotes the estimated latent state and θ_t denotes the corresponding ground-truth value. RMSE provides an aggregate measure of accuracy but does not distinguish between localized and distributed estimation errors.

5.2 Probe Disagreement

To assess the effectiveness of calibration, we measure probe disagreement as the mean absolute difference between redundant probes observing the same latent state. For a given sensor node and modality, probe disagreement is defined as

$$D = \frac{1}{T} \sum_{t=1}^T |y_t^{(1)} - y_t^{(2)}|, \quad (2)$$

where $y_t^{(1)}$ and $y_t^{(2)}$ denote the measurements from two redundant probes at time t . This metric captures internal consistency among probes and is used to verify whether calibration successfully aligns redundant measurements. However, probe agreement alone does not guarantee correctness of the estimated latent state.

5.3 Localized Fault Penalty

Global accuracy metrics such as RMSE may obscure localized failures when faults affect only a subset of sensor nodes or time intervals. To explicitly quantify robustness under sensor faults, we introduce the *localized fault penalty*, defined as

$$\Delta_{\text{fault}} = \text{RMSE}_{\text{fault window}} - \text{RMSE}_{\text{clean window}}. \quad (3)$$

This metric is computed per sensor node and modality by comparing estimation error during a predefined fault window against error during fault-free periods. A positive localized fault penalty indicates degradation under fault conditions, while a negative value indicates reduced error during the fault window.

The localized fault penalty enables targeted analysis of estimator behavior under controlled sensor failures and reveals robustness issues that may be masked by global averaging across nodes or time.

6 Fault Injection Strategy

To systematically evaluate estimator robustness, we introduce controlled sensor faults at the probe measurement level. Rather than attempting to detect or predict faults, our objective is to analyze how different estimation strategies behave when sensor readings become unreliable while the underlying physical process remains unchanged. All faults are injected after data generation and calibration parameter estimation, ensuring that they represent unforeseen sensor failures rather than modeling artifacts.

6.1 Motivation

In real-world sensor networks, failures are often structured and localized, arising from probe degradation, wiring issues, partial sensor malfunction, or environmental interference. Such faults typically manifest as persistent biases or inconsistent readings among redundant probes, rather than as independent random noise. Controlled fault injection provides a principled mechanism to expose these failure modes and assess estimator robustness under well-defined conditions.

6.2 Benign Fault Model

The benign fault model simulates gradual or systematic sensor degradation. A constant bias is added to a single probe of a selected sensor node over a limited time window, while all other probes and nodes remain unaffected. This fault model preserves internal probe consistency and represents common issues such as sensor drift or calibration offset. By construction, the latent ground truth is unchanged, allowing direct assessment of how estimation error responds to biased measurements.

6.3 Adversarial Fault Model

The adversarial fault model is designed to explicitly violate the probe consistency assumption. For a selected sensor node and time window, opposing perturbations are applied to its redundant probes: one probe is biased positively while the other is biased negatively. This construction forces strong disagreement between probes observing the same latent state, mimicking partial probe failure or conflicting sensor signals. As with the benign fault model, the latent ground truth remains unchanged.

6.4 Scope and Assumptions

Faults are injected only at the probe measurement level and do not alter the latent physical states, network topology, or calibration parameters. All estimators are evaluated with full knowledge of the fault time windows for analysis purposes; no attempt is made to detect, localize, or predict faults. The focus of this work is therefore on passive robustness analysis rather than active fault tolerance or diagnosis.

This fault injection framework enables controlled comparison of estimator behavior under nominal conditions, benign sensor failures, and structured adversarial faults, providing clear insight into the strengths and limitations of each estimation strategy.

7 Evaluation Metrics

To evaluate estimator performance under nominal and faulty conditions, we employ a set of complementary metrics that quantify both estimation accuracy and robustness. All metrics are computed with respect to latent ground-truth states, which are used exclusively for evaluation and never exposed to the estimators.

7.1 Root Mean Squared Error

Root Mean Squared Error (RMSE) is used as the primary measure of estimation accuracy under nominal conditions. For a given modality and sensor node, RMSE is computed as

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{\theta}_t - \theta_t)^2}, \quad (4)$$

where $\hat{\theta}_t$ denotes the estimated latent state and θ_t denotes the corresponding ground-truth value. RMSE provides an aggregate measure of accuracy but does not distinguish between localized and distributed estimation errors.

7.2 Probe Disagreement

To assess the effectiveness of calibration, we measure probe disagreement as the mean absolute difference between redundant probes observing the same latent state. For a given sensor node and modality, probe disagreement is defined as

$$D = \frac{1}{T} \sum_{t=1}^T |y_t^{(1)} - y_t^{(2)}|, \quad (5)$$

where $y_t^{(1)}$ and $y_t^{(2)}$ denote the measurements from two redundant probes at time t . This metric captures internal consistency among probes and is used to verify whether calibration successfully aligns redundant measurements. However, probe agreement alone does not guarantee correctness of the estimated latent state.

7.3 Localized Fault Penalty

Global accuracy metrics such as RMSE may obscure localized failures when faults affect only a subset of sensor nodes or time intervals. To explicitly quantify robustness under sensor faults, we introduce the *localized fault penalty*, defined as

$$\Delta_{\text{fault}} = \text{RMSE}_{\text{fault window}} - \text{RMSE}_{\text{clean window}}. \quad (6)$$

This metric is computed per sensor node and modality by comparing estimation error during a predefined fault window against error during fault-free periods. A positive localized fault penalty indicates degradation under fault conditions, while a negative value indicates reduced error during the fault window.

The localized fault penalty enables targeted analysis of estimator behavior under controlled sensor failures and reveals robustness issues that may be masked by global averaging across nodes or time.

8 Graph-Based Robustness Extension

The adversarial fault experiments reveal a key limitation of calibration-only estimation: enforcing probe consistency in isolation can amplify structured sensor failures. To address this limitation, we introduce a deterministic graph-based temporal smoothing extension that incorporates spatial context from neighboring sensors while preserving the benefits of calibration.

8.1 Motivation

Sensor nodes in physical environments are often spatially correlated, as they observe related environmental conditions. When a localized sensor exhibits anomalous behavior, neighboring sensors may provide weak but informative context about the expected latent state. Incorporating such spatial information can regularize estimation and suppress localized fault amplification. However, excessive reliance on neighboring measurements risks introducing bias, particularly under asynchronous sampling.

The objective of the graph-based extension is therefore not to replace calibration, but to act as a conservative regularizer that stabilizes calibrated estimates under structured faults.

8.2 Graph Construction

We represent the sensor network as a graph whose nodes correspond to sensor locations. In the absence of explicit spatial coordinates, we adopt a fully connected graph with self-loops and apply symmetric degree normalization. The resulting normalized adjacency matrix ensures equal weighting across nodes while preserving numerical stability.

Importantly, the graph structure is fixed and does not depend on data-driven learning or optimization. This design choice isolates the effect of spatial coupling from model complexity and learning dynamics.

8.3 Graph-Temporal Estimator

Let $\hat{\theta}_{\text{calib},t}^{(i)}$ denote the calibration-only estimate for sensor node i at time t . The graph-based temporal estimator computes a smoothed estimate as

$$\hat{\theta}_t^{(i)} = (1 - \lambda) \hat{\theta}_{\text{calib},t}^{(i)} + \lambda \sum_j A_{ij} \hat{\theta}_{t-1}^{(j)}, \quad (7)$$

where A_{ij} denotes the normalized adjacency matrix and $\lambda \in [0, 1]$ controls the strength of spatial coupling. The estimator combines the current calibrated estimate with a spatially averaged estimate from neighboring nodes at the previous time step.

This formulation preserves temporal causality and allows spatial information to influence the estimate only through past observations. No additional parameters are learned, and the estimator operates deterministically.

8.4 Sensitivity to Spatial Coupling

The behavior of the graph-based estimator is highly sensitive to the choice of λ . Large values of λ cause the estimate to be dominated by neighbor states, which may be temporally misaligned due to asynchronous sampling. This can introduce bias and increase estimation error under fault conditions.

Conversely, small values of λ apply spatial smoothing conservatively, allowing calibration-only estimates to dominate while suppressing extreme deviations caused by localized adversarial faults. Empirical results demonstrate that values of λ in the range $[0.05, 0.15]$ effectively reduce localized fault penalties relative to the calibration-only estimator.

These findings indicate that spatial context improves robustness only when applied with restraint and that naïve diffusion can be detrimental in asynchronous sensor networks. Figure 3 illustrates the sensitivity of the graph-based estimator to the spatial coupling parameter λ . Small values of λ provide robustness by weakly regularizing calibrated estimates using spatial context, while larger values introduce bias due to asynchronous neighbor dynamics. These results indicate that spatial information improves robustness only when applied conservatively.

8.5 Scope and Limitations

The proposed graph-based extension is intentionally simple and does not adapt its spatial weighting based on data or fault indicators. The adjacency matrix and coupling parameter λ are fixed a priori. While this design enables transparent analysis of spatial effects, it does not capture heterogeneous relationships among sensors or dynamically adjust to evolving fault patterns.

These limitations motivate future extensions involving adaptive or learned spatial weighting, as well as integration with explicit fault detection mechanisms.

9 Experimental Results

We evaluate estimator performance under nominal conditions, benign sensor faults, and adversarial probe-level faults. All results are reported using the metrics defined in Section 5, with particular emphasis on the localized fault penalty to assess robustness. Unless otherwise stated, calibration parameters are estimated once and held fixed for all experiments.

9.1 Nominal Conditions

Under nominal (fault-free) conditions, the calibration-only estimator significantly reduces probe disagreement across all sensor nodes. Redundant probes that exhibit substantial disagreement prior to calibration are closely aligned after calibration, indicating successful recovery of relative gain and bias parameters. Despite this reduction in internal inconsistency, overall RMSE remains comparable to that of the baseline estimator. Figure 1 illustrates the behavior of redundant moisture probes for a representative sensor node under nominal operating conditions. Despite observing the same latent state, the two probes exhibit substantial disagreement due to unknown gain and bias parameters. This discrepancy motivates the need for calibration and highlights that raw probe measurements cannot be directly averaged without introducing bias.

These results confirm that the calibration-only estimator effectively enforces probe consistency without introducing significant bias under nominal operating conditions. They also establish a baseline against which fault-induced degradation can be assessed.

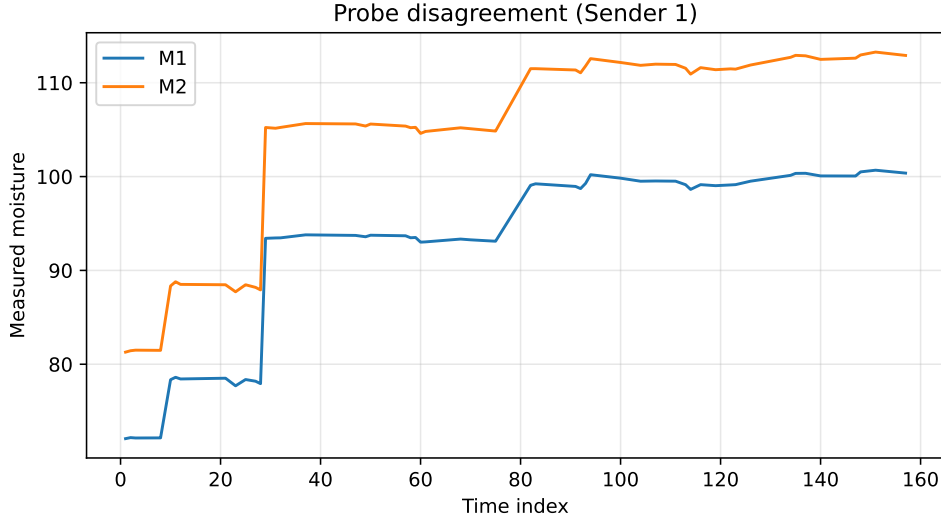


Figure 1: Moisture probe measurements (M1 and M2) for a representative sensor node under nominal conditions, illustrating substantial disagreement prior to calibration.

Table 1 reports RMSE for baseline and calibration-only estimators under fault-free conditions. While calibration significantly reduces probe disagreement, overall RMSE remains comparable to the baseline estimator. This result confirms that calibration primarily enforces internal consistency rather than directly improving global accuracy.

Table 1: RMSE under nominal (fault-free) conditions.

Estimator	Moisture RMSE	Temperature RMSE
Baseline	3.37	0.43
Calibration-only	3.60	0.42

9.2 Benign Fault Results

We first evaluate estimator behavior under benign faults, modeled as constant bias perturbations applied to a single probe over a limited time window. Under this fault model, the calibration-only estimator does not exhibit significant degradation and, in some cases, achieves lower RMSE during the fault window than during fault-free periods.

This counterintuitive behavior arises from accidental cancellation between the injected fault bias and the original calibration bias. While such error reduction may appear beneficial, it does not reflect inherent robustness. Rather, it highlights that calibration can mask certain fault types without explicitly detecting or correcting them. These results demonstrate that reduced estimation error alone is insufficient to characterize robustness.

9.3 Adversarial Fault Results

We next consider adversarial faults designed to explicitly violate the probe consistency assumption. In this setting, opposing perturbations are applied to redundant probes of the same sensor, forcing strong disagreement while keeping the latent ground truth unchanged. Figure 2 compares estimator behavior under adversarial probe-level faults for sensor node 2. The baseline estimator exhibits moderate robustness due to accidental cancellation between opposing probe errors. In contrast, the calibration-only estimator amplifies structured probe disagreement, driving the latent estimate away from the ground truth during the fault window. The graph-based estimator with conservative spatial coupling suppresses this amplification and closely tracks the true latent state.

Under adversarial faults, the calibration-only estimator exhibits clear failure, with localized fault penalties becoming positive for the affected sensor node. By enforcing probe consistency, calibration amplifies the structured disagreement and drives the latent estimate away from the true state. In contrast, the baseline estimator may occasionally exhibit negative fault penalties due to accidental cancellation of opposing probe errors, despite lacking any explicit fault-awareness.

These results demonstrate that probe consistency does not imply correctness and that calibration alone can catastrophically fail under structured sensor faults.

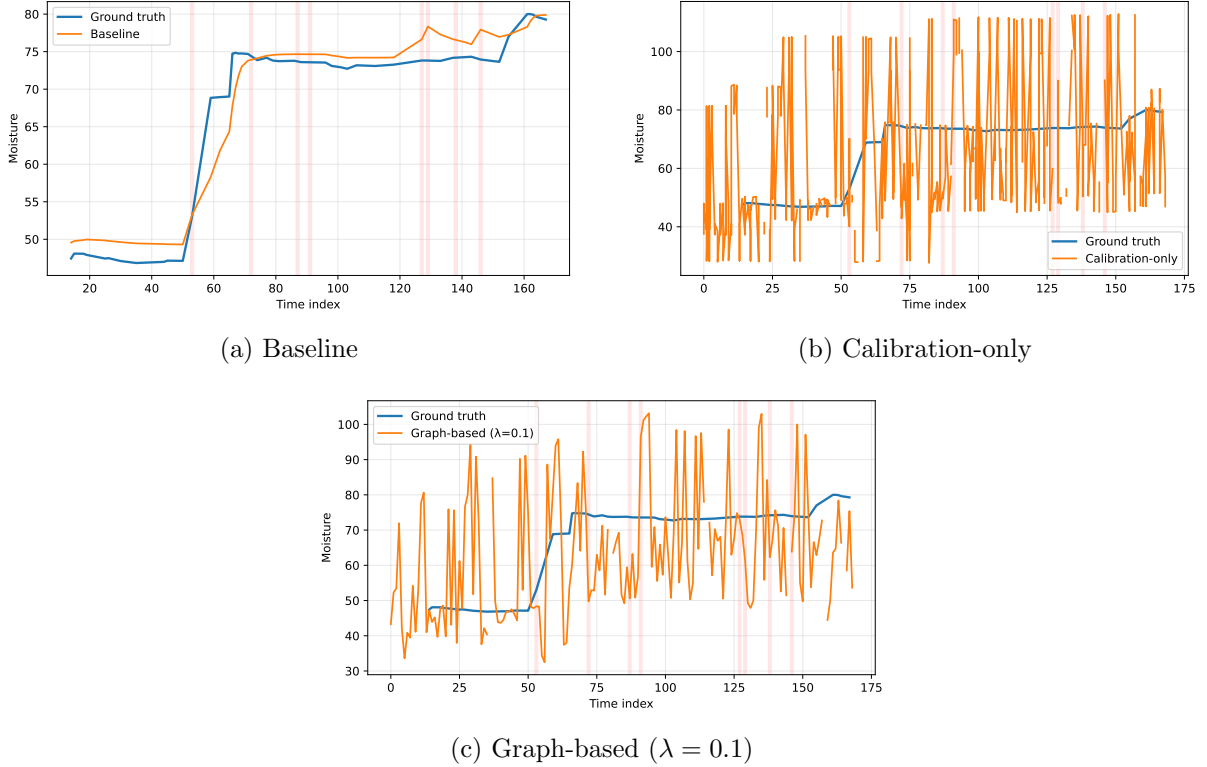


Figure 2: Estimator behavior under adversarial probe-level faults for sensor node 2. The shaded region denotes the fault window.

Table 2 summarizes localized fault penalties under adversarial probe-level faults. Positive penalties indicate degradation during fault windows, while negative values indicate improved robustness. Calibration-only estimation incurs a positive fault penalty, confirming its failure under adversarial conditions. In contrast, conservative graph-based smoothing achieves negative penalties, demonstrating improved robustness relative to both baseline and calibration-only estimators.

Table 2: Localized fault penalty under adversarial probe-level faults (sensor node 2).

Estimator	Localized Fault Penalty
Baseline	−1.90
Calibration-only	+0.43
Graph-based ($\lambda = 0.05$)	−2.95
Graph-based ($\lambda = 0.10$)	−2.28
Graph-based ($\lambda = 0.15$)	−1.38

9.4 Graph-Based Robustness Extension

To mitigate the failure of calibration-only estimation under adversarial faults, we introduce a graph-based temporal smoothing extension that incorporates spatial context from neighboring sensors. The estimator combines calibration-only estimates with spatially averaged neighbor states using a fixed normalized adjacency matrix and a tunable coupling parameter λ .

Experimental results show that estimator robustness is highly sensitive to the choice of λ . Large values of λ introduce bias by overly trusting asynchronous neighbor states, leading to increased localized fault penalties. In contrast, small values of λ act as a weak regularizer that suppresses adversarial fault amplification without significantly biasing the estimate.

For λ in the range $[0.05, 0.15]$, the graph-based estimator achieves substantially reduced localized fault penalties compared to the calibration-only estimator and matches or outperforms the baseline estimator under adversarial fault conditions. These results indicate that conservative spatial coupling can restore robustness while preserving the benefits of calibration.

Overall, the experimental results highlight the nuanced role of calibration and spatial information in fault-prone sensor networks and underscore the importance of localized evaluation metrics for robustness analysis.

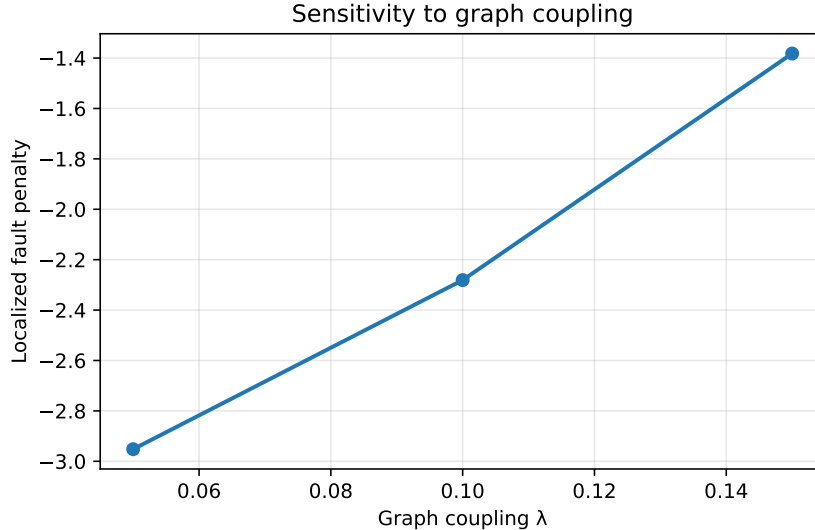


Figure 3: Localized fault penalty as a function of graph coupling parameter λ . Small values provide robustness, while large values introduce bias.

10 Discussion

This work provides a controlled experimental analysis of robust state estimation in redundant sensor networks under structured sensor faults. By isolating calibration effects, fault structure, and spatial coupling, we identify when commonly used estimation strategies succeed and when they fail.

A central finding is that calibration, while highly effective at enforcing internal probe consistency under nominal conditions, does not guarantee robustness. In the presence of benign faults, calibration may reduce estimation error through accidental cancellation of biases, masking underlying sensor degradation. Under adversarial probe-level faults, however, calibration-only estimation can catastrophically fail by amplifying structured disagreement among redundant probes. These results demonstrate that probe consistency should not be conflated with correctness of the latent state estimate.

The introduction of the localized fault penalty proved essential for revealing these failure modes. Global accuracy metrics such as RMSE may obscure localized degradation when faults affect only a subset of sensors or time intervals. By explicitly comparing estimation error during fault windows against fault-free periods on a per-node basis, the localized fault penalty enables targeted robustness analysis and exposes estimator behavior that would otherwise remain hidden.

To address the failure of calibration-only estimation under adversarial faults, we introduced a deterministic graph-based temporal smoothing extension. Experimental results show that conservative spatial coupling can suppress fault amplification and restore robustness when applied with restraint. However, overly aggressive diffusion introduces bias due to asynchronous neighbor dynamics, highlighting a fundamental trade-off between variance reduction and bias

introduction. These findings indicate that spatial context is beneficial only when its influence is carefully controlled.

The proposed graph-based extension is intentionally simple and does not adapt to changing conditions or fault patterns. The adjacency structure and coupling strength are fixed a priori, and no explicit fault awareness is incorporated. While this design enables transparent analysis of spatial effects, it also limits the estimator’s ability to handle heterogeneous sensor relationships or dynamically evolving faults.

These limitations naturally motivate several directions for future work. First, adaptive or learned spatial weighting mechanisms, such as graph neural networks or attention-based diffusion models, may provide a principled means to balance spatial influence without manual tuning. Second, residuals arising from calibration disagreement or graph-based inconsistency offer a natural signal for explicit fault detection and localization. Integrating detection with robust estimation would enable active fault tolerance rather than passive robustness analysis.

Overall, this study establishes a clear empirical foundation for understanding the interaction between calibration, fault structure, and spatial reasoning in sensor networks. By identifying failure modes before introducing learning-based complexity, the results provide a principled baseline upon which more sophisticated robust estimation and fault-aware models can be developed.

11 Conclusion

This report presents a controlled analysis of estimator robustness under structured sensor faults. By isolating calibration and spatial effects, we identify key failure modes and demonstrate how conservative spatial reasoning can mitigate them. These findings provide a principled foundation for future learning-based and fault-detection extensions.